Role of Artificial Intelligence in FACTS-Assisted Power Quality Enhancement: A Comprehensive Review

Abstract *– In modern power systems grid, the main problem is with the quality of Power Quality (PQ) due to some issues like harmonics, voltage sags, and poor power factor. These kinds of problems happens because of proliferation of non-linear loads (e.g., renewable energies). Recent studies have shown that the implementation of Flexible AC Transmission Systems (FACTS) devices have emerged an effective hardware solution for mitigating the mentioned problem via dynamic control of voltage and reactive power. In the meantime, for enhancing the effectiveness of these techniques, Artificial Intelligence (AI) has been increasingly applied to be used alongside FACTS for improving the PQ. This comprehensive review work is summary of the importance of FACTS devices and the utilization of AI in improving the power factor beyond conventional control methodologies. Vital applications of AI – such as intelligence harmonic mitigation, voltage sag prediction with fast corrective actions, and adaptive reactive power compensation – are highlighted. Various AI algorithms (including neural networks, deep learning, and reinforcement learning) enable the control of FACTS devices (like, STATCOM, SVC, and UPFC) to respond better for the manipulation of PQ in different problematic scenarios. This paper provides an in-depth discussion of how AI enhances harmonic filtering, anticipate and mitigates voltage sags, and improves the dynamic of power factor correction. This synthesis establishes a framework for understanding how AI-enhanced FACTS devices can transform from reactive compensators to proactive guardians of power quality in modern electrical networks. Alongside with that, challenges and future direction, for having a better understanding related to the synergy of AI and FACTS offers a promising pathway for superior power quality in smart grids.*

Keywords *- Artificial intelligence, deep learning, flexible AC transmission systems (FACTS), harmonics mitigation, power quality, reinforcement learning, static synchronous compensator (STATCOM), voltage sag compensation.*

1. Introduction

1.1 Power Quality Challenges in Modern Grids

Due to increasing number of non-linear loads, distributed generation (DG) and sensitive electronics, the Power Quality (PQ) has become the major concern in modern electrical grids [1]. Issues like waveform distortion and voltage instability could are prominent problems that can significantly disrupt industrial equipment and operational reliability. For instance, the semiconductor industry can incur production losses estimated $90,000 to $1 million per hour due to the issues related to PQ. Also, The Electric Power Research Institute (EPRI) estimates that power quality disturbances cost U.S. businesses between $119 billion and $188 billion annually [2]. IEEE Standard 1159-2019 categorizes power quality phenomena according to the magnitude, duration, and spectral characteristics, with the establishment of measurement protocols and acceptance thresholds for a diverse operating environment [3]. Renewable energy sources also exacerbate power quality challenges due to the harmonic injection from inverters, voltage fluctuations during rapid resources variations, and phase imbalance in distribution systems [4]. Modern microgrids exhibit particular vulnerability to these disturbances, experiencing rapid voltage deviations during cloud transients in solar-dominated systems that can disrupt sensitive electronic equipment and processes [5].

1.2 FACTS Devices, Principles and Capabilities

For addressing the caused problems, Flexible AC Transmission Systems (FACTS) devices play a critical role in power quality and stability enhancement. These controllers are power-electronics based equipment that are capable of dynamically controlling line parameters like, voltage, impedance, and phase angle for the better improvement of system [6]. With rapid and dynamic injection/absorption of reactive power, FACTS can stabilize voltages, increase transmission capability, and also dampen the disturbances. Other benefits of FACTS according to IEEE [7] are regulated load flow, reduced generation costs, improved stability margins, higher line liability up to thermal limits, and lower reactive power losses. There are various types of FACTS and the key technologies include shunt compensators like the Static Var Compensator (SVC) and Static Synchronous Compensator (STATCOM), series compensators like the Thyristor-Controlled Series Capacitors (TCSC) and combined devices such as the Unified Power Flow Controller (UPFC) and Interline Power Flow Controller (IPFC). These devices have been widely used for the enhancement of voltage profile and overall stability. For instance, STATCOM is a fast-response shunt compensator that could inject or absorb reactive current for the regulation of voltage and power factor; it is also known as its beneficial role like reactive harmonic filtering and flicker mitigation.

1.3 Limitation of Conventional FACTS Control

Convential FACTS control methodologies have high reliability over linear control theory, like proportional-integral (PI) regulators with considering fixed parameters that are derived from small-signum analysis or empirical tuning [8]. While these approaches are functional in compensating steady-state conditions, exhibit significant limitations against the modern power systems. There are three principal limitations characterize conventional control approaches. First, the lack of predictability, that make FACTS to perform as a reactive compensator rather than anticipatory regulator, resulting in delayed response to emerging disturbances [9]. Second, the exhibition of limited adaptability to changing conditions, requiring manual returning as network configurations and load characteristics evolve. Third, multi objective optimization, particularly when the power quality goals conflict with other operational objectives [9].

1.4 AI Integration with FACTS: Transformative Potential

Artificial Intelligence (AI) has already emerged as a key enabler for the optimization of FACTS in real time. AI techniques like neural networks, fuzzy logic, and evolutionary can provide FACTS with adaptive and predictive control capabilities. "Intelligent" FACTS controllers that use AI have more capability of adapting themselves and representing the state-of-art in the control of various situations [10][11]. AI can learn from the historical data alongside with the through heuristic optimization and adjust FACTS settings to mitigate PQ issues more efficiently than conventional ones. The AI implementation in FACTS aligns with the development of the smart grid, wherein advanced control and automation are required for managing the complexity that happens. Recent studies have demonstrated that AI-enhanced FACTS devices can significantly improve power quality metrics. For instance, AI applications in power quality issues have shown potential in reducing harmonic distortion and effectively predicting and mitigating voltage sags. These capabilities are particularly valuable in microgrids and renewable-rich networks, where conventional control approaches often struggle with rapid fluctuations and bidirectional power flows [12].

1.5 Research Gaps and Paper Contributions

Despite promising advances in AI-FACTS integration, research gaps remain in understanding the optimal AI architectures for specific PQ application, the quantification in the performance improvements across diverse network configurations and addressing implementation challenges. This comprehensive review addresses these gaps through four principal contributions:

- Systematic classification of AI techniques applicable for FACTS-assisted PQ enhancement, establishing a taxonomy based on both control objectives and methodological approaches.
- Quantitative assessment of the performance of AI-FACTS in comparison to the conventional control methods.
- Analyzing of implementation challenge including computational requirements, model robustness, and integration with the existence power grid information
- Identifying of merging research directions and technological trends in the AI-driven approach for the PQ management for the future grid modernization efforts

1.6 Paper Organization

This paper contains a comprehensive review of the role of AI in FACTS-assisted power quality enhancement. Section 2 offers an overview of common power quality issues and how using FACTS devices could address them. Section 3 discusses AI-based harmonic detection and filtering by the usage of FACTS. In Section 4, AI techniques for the voltage sag prediction and mitigation via FACTS are examined. Section 5 covers AI-driven dynamic power factor correction and reactive power compensation. In Section 6, the implementation of real-time PQ monitoring and distributed control using AI (in conjunction with IoT and edge computing) is explored. Section 7 highlights the challenges, research gaps, and future trends in this domain. Finally, Section 8 concludes the paper with key insights and implications for future grid modernization efforts.

2. Overview of Power Quality Issues and FACTS Solutions

2.1 Classification of Power Quality Disturbances

The definition of PQ is the set of electrical characteristics that allow electrical equipment to have their proper functionality without facing any significant loss of performance or life expectancy. This definition derived from the IEEE Standard 1159-2019 [3]. Due to penetration of sensitive electronic loads and non-linear power electronic devices, power quality has faced quality concerns that have become increasingly prominent in the modern electrical systems. In continuous part, various power quality disturbances will be discussed.

2.1.1 Voltage Magnitude Variations

2.1.1.1 Voltage Sags(Dips)

Voltage sags are the disturbances by a decrease in Root Mean Squared (RMS) voltage between 0.1 to 0.9 per-unit for the duration of 0.5 cycles to 1 minute. Industry data indicates that voltage sag represents approximately 70% of all power quality disturbances in industrial facilities [13]. Sags are typically classified by their both magnitude and duration, with severity increasing as both parameters increase.

The primary cause of this kind of disturbance include:

- System faults (line-to-line, line-to-ground, etc)
- Large motor starting
- Transformer energization
- Heavy load switching

The mathematical characterization of voltage sag is:

$$S_{sag} = V_{actual} / V_{nominal} \quad (1)$$

S_{sag} is the sag voltage (between 0 and 1)
V_{actual} is the actual voltage level measured during the sag event
$V_{nominal}$ is the standard operating voltage of electrical system

2.1.1.2 Voltage Swells

This phenomenon is a temporarily increase in RMS voltage between 1.1 and 1.8 per-unit lasting from 0.5 cycles to 1minute [14]. The formula related to this fault is as the same as voltage sag one. They occur less frequently than sags but can be equally damaging or equivalent. Common causes include:

- Single line-to-ground faults (on unfaulted phases)
- Load rejection
- Capacitor bank energization
- Incorrect tap settings for transformers

2.1.1.3 Long-Duration Voltage Variations

The long lasting voltage abnormality (less than 0.9 per-unit or bigger than 1.1 per-nit) that takes more than 1 minute, considers as long duration voltage variations. The reason is because of load variations, switching operations, and poor system voltage regulation rather than fault conditions.

2.1.1.4 Interruptions

The complete loss of voltage (under 0.1 per-unit) is called interruption. Based on the amount of time that it takes for being recovered, this error is classified into three categories:

- Momentary: 0.5 cycles to 3 seconds
- Temporary: 3 seconds to 1 minute
- Sustained: exceeding 1 minute

These kinds of error are quantified by their frequency, duration, and severity using various metrics such as System Average Interruption Frequency Index (SAIFI) and System Average Interruption Duration Index (SAIDI) [15].

2.1.2 Waveform Distortion

2.1.2.1 Harmonics

The sinusoidal components of a periodic waveform with frequencies that are integer multiples of the fundamental frequency is called harmonics [16]. By the proliferation of non-linear loads and penetration of renewable energies, harmonic distortion has been considered as a pervasive issue in the modern power grid system. The IEEE 519-2014 as considered limits for harmonic voltage and current distortion at the Point of Common Coupling (PCC) with having more stringent requirements for higher voltage levels and various conditions. The quantification of this disturbance is based on the following formula that calls Total Harmonic Distortion (THD):

$$THD = \sqrt{(\sum(V_h)^2 / V_1^2)} \quad (2)$$

V_h is the RMS voltage of harmonic order h
V_1 is the fundamental voltage

Another quantified method is for the Individual Harmonic Distortion (IHD) and it is calculated:

$$IHD_h = V_h / V_1 \quad (3)$$

The main reasons of having this issue are:

- Switch-mode power supplies
- Variable frequency drives
- LED lighting
- Arc furnaces
- Saturated transformers
- Battery chargers

2.1.2.2 Interharmonics

Frequency components that are not integer multiples of the fundamental frequency are called interharmonics [17]. They are increasingly prevalent due to:

- Cycloconverters
- Arcing devices
- Induction motors with varying loads
- Power electronic converters with asynchronous switching

Interharmonics can cause:

- Light flicker
- Thermal stress in capacitor banks
- Interference with control and protection signals
- Disturbance in ripple control systems

2.1.2.2 Notching

This occurrence represents periodic voltage disturbances caused by then normal operation of power electronic devices in the time of current commutating from one phase to another [18]. It severity characterized by:

- Notch depth: The voltage deviation during the notch
- Notch area: The time-voltage product of the notch
- Notch count: The frequency of occurrence

2.1.2.3 Noise

Signal waveform with the frequencies of less than 200kHz superimposed on the power system voltage or current is considered as noise [19] and it categorized as:

- Common mode (between conductors and ground)
- Transverse mode (between line conductors)

2.1.3 Voltage Fluctuations and Flicker

Voltage fluctuations refer to the random changes in the voltage typically between 0.9 to 1.1 per-unit. These changes in voltage often occur in the frequency range of 0.5-30 Hz that can cause visible flicker as well [20]. The IEC 61000-3-7 [21] and IEEE 1453-2015 [22] standards establish planning levels and compatibility levels for flicker severity. Flicker severity is quantified using:

Short-term flicker severity (P_{st}): Measured over a 10-minute period
Long-term flicker severity (P_{lt}): Measured over a 2-hour period

$$P_{st} = \sqrt{(\sum w_i P_i^2 / \sum w_i)} \quad (4)$$

$$P_{lt} = \sqrt[3]{(\sum P_{st}^3 / N)} \quad (5)$$

w_i is the weighting coefficient
P_i is the percentile level of the flicker sensation

Primary sources of voltage fluctuations include:

- Arc furnaces
- Welding equipment
- Rolling mills
- Reciprocating pumps and compressors
- Wind turbines
- Large motor starting (repetitive)

2.1.4 Power System Imbalance

2.1.4.1 Voltage Unbalance

In the normal scenario of three-phase voltage system, the magnitude of waves should be the same with difference of 120 digress in their phases. However, due to abnormal conditions in the grid an unbalance voltage could happen. Typical industry standards limit voltage unbalance to 2-3% [21] [21]. This disturbance could be quantified using the following equation:

$$\%VU = |V_2| / |V_1| \times 100 \quad (6)$$

V_1 and V_2 are the positive and negative sequence voltage components.

This phenomenon primary results from:

- Uneven distribution of single-phase loads across three phases
- Open delta transformer connections
- Blown fuses in capacitor banks
- Asymmetrical transformer impedances
- Untransposed transmission lines

2.1.4.2 Current Unbalance

Similar to the voltage unbalance, it defined as [21]:

$$\%CU = |I_2| / |I_1| \times 100 \quad (7)$$

I_1 and I_2 are the positive and negative sequence current components.

It is important to note that the unbalance condition in the current system is distinguish from the voltage and in the balanced voltage, this occurrence could appear.

2.1.5 Power Frequency Variations

Any deviation from the nominal system frequency (50/60 Hz) more than the nominal limitation, represents another category of power quality disturbance [22]. These are categorized as:

2.1.5.1 Frequency Deviations

It happens due to the imbalanced condition among generation and load with [22]:

- Normal range: 0.1 Hz in interconnected systems
- Acceptable range: 0.5 Hz during disturbance

2.1.5.2 Frequency Variations

It includes both steady-state variations in isolated systems and dynamic variations during system disturbances [24]:

- Rate of Change of Frequency (ROCOF): Critical for islanding detection
- Frequency Nadir: Minimum frequency reached during a disturbance
- Frequency Recovery Time: Time to return to acceptable limits

For the categorization of the characteristics of frequency performance, statistical terminology like mean, standard deviation, and cumulative frequency distributions are used.

2.1.6 Transient Disturbances

2.1.6.1 Impulsive Transients

Sudden disturbances that are non-power frequency changes in voltage or current that have unidirectional characteristic in polarity are considered as impulsive transients [23]. Primary caused include lightning strikes, switching of inductive loads, and electrostatic discharge. These disturbances are characterized by:

- Rise time: 1-10 μs
- Duration: <50 μs
- Magnitude: Up to several kV

2.1.6.2 Oscillatory Transients

Like the previous disturbance, with the difference of impact on the both positive and negative polarity values of current or voltage, is called the oscillatory transient one. The severity of this issue is quantified by peak magnitude, frequency content, and energy content. It also be classified by frequency range:

- Low frequency: <5 kHz (capacitor switching)
- Medium frequency: 5-500 kHz (line/cable switching)
- High frequency: 500 kHz-5 MHz (lightning, switching in isolated systems)

2.1.7 Power Factor and Reactive Power Issues

Low power factor has high negative impact over the both power quality and system efficiency [25] [26]. These issues caused by various reasons:

- Increased current for the same active power transfer
- Higher voltage drop along lines
- Reduced equipment capacity utilization
- Increased system losses proportional to the square of current

For both sinusoidal (Power Factor, PF) and non-sinusoidal (Displacement Power Factor, DPF) conditions this occurrence can be quantified using below formulas:

$$PF = P / S = P / (V_{rms} \times I_{rms}) \quad (8)$$

$$True\ PF = DPF \times (1 / \sqrt{(1 + THD_i^2)}) \quad (9)$$

$$S = \sqrt{(P^2 + Q^2)} \quad (10)$$

PF is the power factor
True PF is the actual power of the system with the consideration of both displacement and distortion components.
P is the active power
Q is the reactive power
S is the apparent power
V_{rms} is the effective voltage
I_{rms} is the effective current
DPF is the displacement power factor and it is related to the phase angle between voltage and current
THD_i is the measure of how much harmonic distortion is present in the current waveform

The mentioned power quality disturbances can be seen at Table 1 based on their types, characteristics, and impacts [27]:

Category	Disturbance Type	Description	Primary Causes
Voltage Magnitude Variations	Voltage Sags (Dips)	RMS voltage drops (0.1-0.9 p.u.) lasting 0.5 cycles to 1 min	System faults, large motor startup, transformer energization
	Voltage Swells	RMS voltage rise (1.1-1.8 p.u.) lasting 0.5 cycles to 1 min	Load rejection, capacitor switching, incorrect transformer tap settings
	Long-Duration Voltage Variations	Voltage fluctuations lasting over 1 min	Load variations, switching operations, poor voltage regulation
	Interruptions	Complete voltage loss (<0.1 p.u.)	Faults, equipment failures, grid disturbances
Waveform Distortion	Harmonics	Integer multiples of fundamental frequency causing distortion	Non-linear loads, inverters, rectifiers, transformers
	Interharmonics	Non-integer frequency distortions	Cycloconverters, arcing devices, power converters
	Notching	Periodic waveform distortions due to commutation in power electronics	Power electronic switches
	Noise	High-frequency electrical noise (<200 kHz) affecting signals	Electromagnetic interference, faulty grounding
Voltage Fluctuations & Flicker	Voltage Fluctuations	Voltage changes (0.9-1.1 p.u.) causing flicker	Arc furnaces, welding, large motor startups
Power System Imbalance	Voltage Unbalance	Unequal phase voltages causing instability	Uneven load distribution, blown fuses, transformer imbalance
	Current Unbalance	Unequal phase currents affecting efficiency	Unbalanced loads, transformer asymmetry
Power Frequency Variations	Frequency Deviations	Variations in nominal frequency (50/60 Hz)	Load-generation imbalances, system faults
	Transient Disturbances	Short-duration high-magnitude voltage events	Lightning, capacitor switching, inductive loads
Power Factor & Reactive Power Issues	Low Power Factor	Inefficient power use increasing losses	Inductive loads, poor compensation

Table 1: Classification of Power Quality Disturbances and Their Causes

2.2 FACTS Devices, Classification and Operating Principles

FACTS devices, represent a sophisticated family of power electronic-based controllers that their main reason of design was to enhance the controllability, stability, and power transfer capability of AC transmission systems [28] [29]. This chapter focuses on various aspects of FACTS devices and how they can be impactful over the increasing the functionality of the grid with the mitigation of the disturbances.

2.2.1 Fundamental Principles and Theoretical Basis

The two main features that are controlled using FACTS that have vital impact over the power quality are Active Power (P) and Reactive Power (Q). The related power equation to these two metrics in a two-bus system are [30]:

$$P = (|V_s| \times |V_r| / X) \times \sin(\delta_s - \delta_r) \quad (11)$$

$$Q_s = (|V_s|^2 / X) - (|V_s| \times |V_r| / X) \times \cos(\delta_s - \delta_r) \quad (12)$$

P is the active power transmission
Q_s is the reactive power at the sending end
$|V_s|$ and $|V_r|$ are the voltage magnitudes at sending and receiving ends
δ_s and δ_r are the voltage phase angles
X is the line reactance

FACTS devices provide dynamic control over the three main parameters of the mentioned equations:

- Line impedance (X)
- Voltage magnitude (|V|)
- Phase angle (δ)

This dynamic control happens using through insertion of reactive elements (like inductors and capacitors) or direct voltage and current injection using the power electronic converters that normally operates at high switching frequencies [25].

2.2.2 Classification Based on Connection Configuration

2.2.2.1 Series-Connected FACTS Controllers

These kinds of FACTS devices are connected in series with the transmission lines through coupling transformers, for controlling the impedance or voltage sources [31]. With this configuration, the direct control of power flow (that is relatively has smaller ratings compared to shunt devices) is possible according to the following relationship:

$$X_{effective} = X_{line} \times (1 - k) \quad (13)$$

$X_{effective}$ is the effective line reactance after series compensation.

2.2.2.1.1 Thyristor-Controlled Series Capacitor (TCSC)

TCSC contains of a fixed number of capacitor that are in parallel with a Thyristor-Controlled Reactor (TCR), forming a controllable capacitive reactance [32] [33]. There are two main stages in its operation:

- Fundamental Circuit: Series capacitor (X_C) in parallel with a TCR comprising an inductor (X_L) and a bidirectional thyristor valve.
- Impedance Characteristics: The effective impedance varies with thyristor firing angle(α) that is range from its minimum (near resonance point) to 90 (fully conducting)

$$X_{TCSC} = X_C \times X_L(\alpha) / (X_L(\alpha) - X_C) \quad (14)$$

$$X_L(\alpha) = X_L \times \pi / (\pi - 2\alpha + \sin 2\alpha) \quad (15)$$

- Operating Modes:
  Bypassed Mode: Thyristor fully conducting(α=90°)
  Blocked Mode: thyristors blocked (equivalent to fixed capacitor)
  Capacitive Vernier Mode: $X_{TCSC} > X_C$ (firing angle in inductive region)
  Inductive Vernier Mode: $X_{TCSC} < 0$ (firing beyond resonance point, limited use)
- Resonance Considerations: A forbidden region exists near the resonance condition that operation in this region is avoided to prevent any high harmonic amplification [34]:

$$X_L(\alpha) = X_C \quad (16)$$

The TCSC has various benefits for the transmission system:

- Power flow control with impedance modulation between 20-70% of the line conductance
- Capability of damping in power oscillation through dynamic impedance control
- Mitigation of Sub Synchronous Resonance (SSR) via appropriate control algorithms
- Limitation of fault current by rapidly increasing the line impedance

2.2.2.1.2 Static Synchronous Series Compensator (SSSC)

This device employs a Voltage-Source Converter (VSC) to inject a controllable voltage in series with the transmission line, usually in quadrature with the line current to emulate a controllable reactance [35] [36] [37]:

- Circuit Configuration: A VSC connected to the transmission line through a series coupling transformer, with a DC capacitor maintaining the DC bus voltage.
- Operating Principle: The SSSC could inject V_q with controllable magnitude and phase. In the following equation, $\theta_q = \theta_i \pm 90$, where θ_i is the phase angle of the line current:

$$V_q = |V_q| \times e^{j\theta_q} \quad (17)$$

  Power Flow Equations:

$$P = (|V_s| \times |V_r| / X) \times \sin(\delta_s - \delta_r) + (|V_s| \times |V_q| / X) \times \sin(\delta_s - \theta_q) \quad (18)$$

$$Q_s = (|V_s|^2 / X) - (|V_s| \times |V_r| / X) \times \cos(\delta_s - \delta_r) - (|V_s| \times |V_q| / X) \times \cos(\delta_s - \theta_q) \quad (19)$$

- Control Capabilities:
  Reactance Emulation: Maintains $\theta_q = \theta_i \pm 90$ with variable $|V_q|$
  Direct Phase Angle Control: Varies θ_q to directly influence power angle
  Combined Control: Simulates control of reactance and phase angle
  Multiline Control: When connected to multiple lines (IPFC configuration)

This specific device offers different benefits:

- Independent control of reactive compensation regardless of line current magnitude
- No resonance concerns
- Smaller physical footprint for equivalent compensation
- Superior dynamic performance during transients
- Capability for active power exchange with appropriate energy source

2.2.2.1.3 Dynamic Voltage Restorer (DVR)

The DVR is categorized as series-connected voltage controller that its responsibility is to control voltage sag/swell mitigation [38] [39]:

- Basic Configuration: It consists of a VSC, series injection transformer, DC energy storage (typically capacitors or supercapacitors), and control system.
- Operating Principle: It can detect voltage disturbances and injects compensating voltage:
- Compensation Strategies:
  In-phase compensation: Where V_{inj} is in phase with pre-sage voltage
  Pre-sag compensation: Restores voth magnitude and phase angle
  Minimum energy compensation: Minimize the active power that is drawn from the storage
  Zero active power compensation: Injects voltage in quadrature with current
- Mathematical Formulation: The required voltage for compensating is calculated using the follow formula:

$$V_{inj} = V_{ref} - V_{actual} \quad (20)$$

$$S_{DVR} = V_{inj} \times I_L{}^* \quad (21)$$

V_{ref} is the desired voltage
V_{actual} is the disturbed supply voltage
S_{DVR} is the apparent power
$I_L{}^*$ is the complex conjugate of load current

DVRs typically provide:

- Sag/swell compensation up to 50% of nominal voltage
- Response times of 10-25 milliseconds
- Protection duration of several cycles to seconds depending on energy storage capacity
- Active filtering capability for upstream harmonic voltages

2.2.2.2 Shunt-Connected FACTS Controllers

Sunt controllers are connected in parallel with the system at specific points to have their most impact on the system (typical locations are at load buses or midpoints of long transmission line) functioning as controllable current source or variable impedance [40].

2.2.2.2.1 Static Var Compensator (SVC)

The SVC is considered as the first-generation of shunt-connected FACTS device that is utilized thyristor-switched or thyristor-controlled reactive elements for providing a dynamic reactive power compensation [41] [42].

- Configuration Components:
  Thyristor-Controlled Reactor (TCR): Variable inductive reactance
  Thyristor-Switched Capacitor (TSC): Discretely switchable capacitive reactance
  Fixed Capacitor (FC): Provides base compensation and harmonic filtering
  Mechanically-Switched Capacitors (MSC): For slower and larger adjustments
- TCR Operation: Reactive power absorption varies with firing angle:

$$Q_{TCR} = (V^2 / \omega L) \times (2 \times (\pi - \alpha) + \sin 2\alpha) / \pi \quad (22)$$

  Q_{TCR} is the reactive power of the TCR
  V is the system voltage
  ω is the angular frequency
  L is the inductance of the reactor
  α is the firing angle
  Control Characteristic: The voltage regulation follows:

$$I_{SVC} = (V_{ref} - V) / X_{SL} \quad (23)$$

  V_{ref} is the reference voltage
  X_{SL} is the slope reactance (typically between 1% to 5%)

The application of SVC includes:

- Voltage regulation within ±3% at load buses
- Reactive power compensation up to several hundred MVAr
- Dynamic stability enhancement through modulation controller
- Flicker mitigation with response times of 2-3 cycles
- Unbalance compensation with phase-independent control

2.2.2.2.2 Static Synchronous Compensator (STATCOM)

This device utilizes a voltage source controller for producing a controllable reactive power through voltage magnitude regulation [43] [44]. This exchange reactive power is expresses as:

$$Q_{STATCOM} = (V_s \times (V_s - V_{STATCOM})) / X \quad (24)$$

$$P_{STATCOM} = (V_s \times V_{STATCOM} \times \sin \delta) / X \quad (25)$$

$Q_{STATCOM}$ is the reactive power exchange
V_s is the system voltage
$V_{STATCOM}$ is the converter output voltage
X is the coupling reactance
δ is the phase angle between voltages
Control Mode
Voltage Regulation Mode: Maintain bus voltage at reference value
Reactive Power Control: Provides specified reactive power
Power Factor Control Mode: Maintains specified power factor
Var Reserve Control Mode: Maintains specified reactive margin
Advanced Control Features
Positive-Negative Sequence Control: For unbalance compensation
Harmonic Compensation: Active filtering up to 13th harmonic
Subsynchronous Damping: Through auxiliary modulation control
Flicker Mitigation: With high-speed reactive power modulation
STATCOM Advantageous
Superior dynamic performance with response times <1 cycle
Full reactive current capability at low voltages
Smaller footprints for equivalent compensation
Lower harmonic generation and filtering requirements
Potential for active power exchange with appropriate energy storage

2.2.2.2.3 Distribution STATCOM (D-STATCOM)

It's a STATCOM variant optimized for distribution systems, typically rated 1-10 MVA [45] [46] [179].

- Circuit Topology: Its similar to the STATCOM but with design adaptions for distribution systems.
  Higher switching frequency in comparison to STATCOM (2-5 kHz)
  Lower voltage ratings (typically less than 35kV connection)
  More compact footprint
  Optional Integrated active power capability
- Mathematical Model:

$$I_{D\text{-}STATCOM} = (V_s \times (V_s - V_{D\text{-}STATCOM})) / X_L \quad (24)$$

$$Q_{D\text{-}STATCOM} = (|V_s| \times |V_{D\text{-}STATCOM}| \times \sin(\delta)) / X_L \quad (25)$$

  $I_{D\text{-}STATCOM}$ is the injected current by the D-STATCOM
  V_s is the system voltage
  X_L is the inductive reactance
  $Q_{D\text{-}STATCOM}$ is the reactive power injected or absorbed by the D-STATCOM
  δ is the phase angle between voltages
  $|V_{D\text{-}STATCOM}|$ is the magnitude of compensator voltage
  $|V_s|$ is the magnitude of system voltage
  Control System:
  Outer voltage regulation loop (response time ~100ms)
  Inner current control loop (response time ~10ms)
  Phase-locked loop for synchronization
  Optional active control loop with energy storage
  D-STATCOM Applications:
  Voltage regulation at weak points
  Flicker mitigation for loads such as arc furnaces
  Power factor correction near large industrial loads
  Voltage sag mitigation
  Harmonic filtering up to the 25th harmonics

2.2.2.4 Combined Series-Shunt Controllers

In the combined controllers, both series and shunt compensators have been emerged, typically connected through a common DC link, enabling comprehensive power flow and voltage control.

2.2.2.4.1 Unified Power Flow Control (UPFC)

UPFC is considered as the most adaptable controller, contains the combination of STATCOM and SSSC with a common DC link [47] [48] [49] [56] [180].

- Circuit Configuration:
  Shunt converter (STATCOM equivalent) connected with transformer
  Series converter (SSSC equivalent) connected with transformer
  Common DC link with capacitor storage
  Control system for multiple control loops
- Operating Principle:
  Line impedance (through series voltage injection)
  Voltage magnitude (through shunt reactive compensation)
  Phase angle (through active power exchange between converters)
- Mathematical Formulation: The series injected voltage is:

$$V_{inj} = \rho \times V_s \times e^{j\gamma} \quad (26)$$

  ρ is the injection coefficient (typically between 0 and 0.5)
  γ is the controllable angle
  Power Transfer Capability: The UPFC can theoretically adjust the power flow from zero to the thermal limit of the line, regardless of the line impedance or phase angle
  Control Modes:
  Power Flow Control Mode: Maintains specified active and reactive power
  Line Impedance Emulation Mode: Simulates variable series reactance
  Phase Angle Regulation Mode: Controls phase angle shift
  Voltage Regulation Mode: Maintains receiving end voltage
  Multifunction Mode: Combinations of above functions
  The UPFC provides unprecedented control capabilities:
  Simultaneous, independent control of active and reactive power flow
  Voltage regulation at the connection point
  Improvement of transient stability through power flow modulation
  Damping of power system oscillations
  Balancing of asymmetric loading

2.2.2.4.2 Unified Power Quality Conditioner (UPQC)

UPQC is a distribution-level device that combines series and shunt active filters for comprehensive power quality enhancement [50] [51] [52] [57] [185] [230].

- Configuration: Similar to UPFC but optimized for power quality applications:
  Shunt converter for current-related issues (harmonics, unbalance, power factor)
  Series converter for voltage-related issues (sags, swells, flicker, harmonics)
  Common DC link with energy storage
  High-speed digital control system
- Mathematical Model: The series and shunt compensating components are:

$$i_{sh} = i_h + i_q + i_u \quad (27)$$

$$v_{se} = v_{ref} - v_s \quad (28)$$

  i_h is the harmonic component
  i_q is the reactive component
  i_u is the unbalanced component of load current
  Control System:
  Series controller: Typically implements instantaneous reactive power theory
  Shunt controller: Uses synchronous reference frame or p-q theory for extraction
  DC link controller: Maintains capacitor voltage
  Synchronization system: Provides reference signals
  Compensation Capabilities:
  Voltage sags/swells: Typically up to 50% for several cycles
  Current harmonics: Up to 25th harmonic with >90% compensation
  Voltage harmonics: Up to 13th harmonic with >85% compensation
  Unbalance: Reduction of negative sequence components by 80-90%
  Flicker: Reduction of flicker severity by 65-80%
  UPQC applications:
  Critical manufacturing facilities with sensitive processes
  Data centers and telecommunications facilities
  Hospitals and healthcare facilities
  Microgrids with high penetration of renewables
  Industrial parks with mixed loads

2.2.2.4.3 Interline Power Flow Controller (IPFC)

The IPFC extends the concept of the SSSC to multiple lines, allowing control of power flow among different transmission corridors. The IPFC is particularly useful in meshed networks where controlling power distribution among multiple paths is critical for congestion management and security enhancement [53] [54] [58].

- Configuration: Multiple series VSCs with a common DC link, each injecting controllable voltage into different lines.
- Operating Principle: Enables active power exchange between lines while providing reactive compensation:

$$\sum(V_i \times I_i{}^*) = 0 \quad (29)$$

  V_i is the injected voltage in line i
  I_i is the current in line i
  Control Capabilities:
  Redistribution of power flow among lines
  Compensation for reactive power demands
  Minimization of overall system losses
  Improvement of system stability through coordinated control

Feature	SVC	STATCOM	TCSC	SSSC	UPFC/UPPQC
Connection	Shunt	Shunt	Series	Series	Combined
Technology Base	Thyristor	VSC	Thyristor	VSC	VSC
Response time	2-3 cycles	<1 cycle	1-2 cycles	<1 cycle	<1 cycle
Control Range	Limited by V²	Independent of V	Limited range	Independent of I	Comprehensive
Maximum Compensating Current	Proportion to V	Rated value	N/A	N/A	Rated value
Maximum Compensating Value	N/A	N/A	Limited to MOV	Rated value	Rated value
Reactive Capability	V²B	Independent of V	N/A	N/A	Comprehensive
Active Power Control	No	With ESS only	Indirect	With ESS only	Direct
Overload Capability	30% for 1s	1.5-2x for 1s	30% for 1s	1.5-2x for 1s	1.5-2x for 1s
Harmonic Generation	2-3% THD	<1% THD	2-3% THD	<1% THD	<1% THD
Harmonic Filtering Capability	Limited	Up to 25th	None	None	Up to 25th
Maximum Practical Rating	800 MVAr	400 MVAr	350 MVAr	200 MVAr	300 MVAr
Response to System Faults	Poor at low V	Excellent	Limited	Good	Excellent
Space Requirements	0.008-0.12 m²/MVAr	0.03-0.05 m²/MVAr	0.06-0.08 m²/MVAr	0.05-0.07 m²/MVAr	0.08-0.12 m²/MVAr
Losses at Rated Output	0.7-1.0%	1.2-1.8%	0.7-1.0%	1.4-2.0%	1.6-2.5%
Life Expectancy	30-40 years	25-30 years	30-40 years	25-30 years	25-30 years
Speed Control Loop Bandwidth	5-10 Hz	100-300 Hz	3-5 Hz	100-300 Hz	100-300 Hz
Relative Cost ($/kVAr)	30-50	50-70	40-60	70-90	90-120
Acoustic Noise	60-80 dB	65-75 dB	55-75 dB	65-75 dB	65-80 dB
Cooling Requirements	Air/Oil	Air/Liquid	Air/Oil	Air/Liquid	Air/Liquid
Control Complexity	Medium	High	Medium	High	Very High
Operating Temperature Range	-40 to +50°C	-35 to +45°C	-40 to +50°C	-35 to +45°C	-35 to +45°C
Altitude Derating (above 1000m)	1%/100m	1%/100m	1%/100m	1%/100m	1%/100m

Table 2: Comparison of FACTS Devices Characteristics [55] [59] [60]

3. Fundamental of Artificial Intelligence in Power Systems

Artificial Intelligence (AI) is a revolutionizing moder power system operations, control and analysis that could help the grid for its resilience and reliability enhancement [61]. This section introduces the fundamental AI technologies relevant to power systems before delving into their specific application for the power grid and harmonic distortion mitigation.

3.1 Core AI Technologies for Power System Applications

AI contains of computational methods that resemble the human-decision making behaviour. Several branches of AI have proven highly effective in power system application.

3.1.1 Machine Learning

Machine Learning (ML) is the heart of AI-driven power system solutions. It aids the system decision making process to enhance its performance on data rather than only on predefined rules [62] [63].

ML algorithms analyze historical operational data for pattern recognition and relationships. The learning process follows:

$$y = f(x; \theta) \quad (30)$$

Here f maps the inputs x (e.g., system measurements) to output y (e.g., load forecasts or equipment status) with the parameter θ optimization through iterative training. This approach can be useful for power system, as it's a physical model and in these kinds of models can be incomplete or computationally demanding.

3.1.2 Neural Networks

It is a key subset of ML, that has wide-ranging applications in power systems [64] [65]. An artificial neuron computes:

$$y = \varphi(\sum w_i x_i + b) \quad (31)$$

x_i is the inputs
w_i is the associate weight to each input
b is the bias
φ is the activation function

In the modern implementation, advanced architectures such as feedforward networks for static predictions and recurrent networks for handling temporal sequences are used [66]. For the further extension, deep learning uses hierarchical features, that also enables more accurate and efficient solutions for complex power challenges.

3.1.3 Fuzzy Logic

In this method of uncertainties management in power systems, digress of truth is utilized rather than bisary logic [67]. Unlike traditional methods, that apply grid classification, fuzzy logic assigns membership values. This method enables more adaptive classification of operating conditions and gives enough control responsibility, reducing instability and improving the operational efficiency:

$$\mu_A(x) : X \to [0,1] \quad (32)$$

A represents the fuzzy test
x represents an element belonging to the universal set X
X is the domain where elements x are drawn from
μ_A is the membership function

3.1.4 Evolutionary and Swarm Intelligence

These algorithms and swarm intelligence techniques use nature-inspired methods to optimize the power system performance [69] [70]. By iteratively refine a population of candidate through selection and variation, they try to find the best outcome. These techniques are effective to address the complex optimization problems that contain non-convex, discontinuous, or mixed-integer search spaces, where traditional methods often struggle to find the best results.

$$f(x) = \min/\max f(x),\ x \in \Omega \quad (33)$$

3.2 AI Applications Across the Power System Value Chain

AI techniques are utilized to transform every stage of the modern power system, from generation to consumption part, with varying levels of implementation maturity.

3.2.1 AI in Power Generation

Due to high penetration of RES and their intermittency characteristics, it is important to have a method that has the role of forecasting to improve the accuracy of RES (like solar and wind) power predictions by integrating weather data, historical patterns, and spatial correlations [71] [72]. Deep learning models can be used for this matter:

$$P_{pv} = f(G, T, C, t, P_{history}) \quad (34)$$

G is the solar irradiance
T is the temperature
C is the cloud coverage

Other AI models also can incorporate atmospheric dynamics and turbine-specific characteristics. That could help enhance the accuracy forecasts, reduce reserve requirements, and lower integration costs.

AI also can be revolutionizing generator condition monitoring and predictive maintenance [73]. With the implementation of convolutional and recurrent neural networks, AI analyze complex sensor data for the early signs of equipment malfunction detection:

$$H(t) = f(S(t), S(t-1), \ldots, S(t-n)) \quad (35)$$

By the help of these models, potential failures can be identified weeks or months in advance, enabling condition-based maintenance that could reduce the unexpected breakdown and unnecessary serving.

3.2.2 AI in Transmission Systems

Ai-driven dynamic line rating prediction optimizes transmission capacity by factoring in environmental condition like ambient temperature, wind speed, and solar irradiation [74].

$$I_{max} = f(T_{amb}, V_{wind}, \theta_{wind}, S_{solar}) \quad (35)$$

This method increases transmission capacity in comparison to traditional seasonal ratings, helping to reduce contention and enable more renewable energy integration without costly physical grid expansion.

3.2.3 AI in Distribution System Management

AI method can improve load forecasting, network reconfigurations, and non-technical loss detection in distribution networks [75] [76]. For instance, Graph Neural Networks (GNNs) effectively model the complex structure of distribution grids with using the following formula:

$$H^{(l+1)} = \sigma(D^{-1/2} A D^{-1/2} H^{(l)} W^{(l)}) \quad (36)$$

A represents the adjacency matrix of the network
$H^{(l)}$ is representing the node features at layer l
D is the matrix degree
$D^{-1/2}AD^{-1/2}$ is the symmetric normalization
$W^{(l)}$ is the trainable weight matrix
σ is the activation function

These models enable topology-aware optimization, incorporating both electrical and physical constraints for the network efficiency improvement.

3.2.4 AI in Distribution System Management

Reinforcement Learning (RL) algorithms are AI methods that used in microgrid management and energy optimization to develop optimal control policies based on system interaction simulations [77].

$$\pi^*(s) = \mathrm{argmax}_a Q^*(s, a) \quad (36)$$

π* represents the optimal policy
$\mathrm{argmax}_a Q^*(s, a)$ is the action that maximize the long-term rewards at given state s

These RL-based controllers help to balance the cost minimization, reliability, and operational constraints while adapting to uncertainties in the grid.

3.3 Advantages of AI in Power System Application

AI techniques offer numerous advantages over conventional methods in different power system domains [78]. Their capability illuminate itself for the complex patterns recognition that allows them to extract valuable insights from high-dimensional operation data. Subtle correlations in grid measurements can be recognized using deep neural networks that traditional statistical approaches may overlook. This feature leads these methods to more accurate forecasting, abnormality detection, and system characterization.

3.3.1 Adaptive Learning and Model Refinement

Adaptability of AI methods is considered as one of the vital benefits that can continuously adjust itself with a new environment based on the changes in the power system conditions [79]. Conventional methods often require manual adjustments in the matter of source generation, load characteristics, or network topologies evolve, where AI models, can learn dynamically from new operational data. The following formula represents the gradient-based optimization update rule [80]:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \quad (37)$$

θ_t is the model parameters at time t
$\nabla L(\theta_t)$ is the gradient of the loss function
η is the learning rate
θ_{t+1} is the updated parameters for next iteration

This adaptive learning is particularly valuable in power grids with the high renewable energy penetration, evolving market mechanisms, and shifting consumer demand, to ensure that the models could remain accurate in any severe grid conditions.

3.3.2 Predictive Capabilities and Proactive Decision-Making

AI can develop further beyond the traditional reactive approaches by anticipating future system conditions, from load variations to equipment failures [81]. Recurrent Neural Networks (RNN), are designed for temporal sequence modeling, which makes the prediction possible based on historical data and current conditions [82]:

$$h_t = \tanh(W_{xh} x_t + W_{hh} h_{t-1} + b_h) \quad (38)$$

$$y_t = W_{hy} h_t + b_y \quad (39)$$

h_t is the hidden state at time t
x_t is the input at time t
W_{xh} is the input-to-Hidden weight matrix
W_{hh} is the hidden-to-hidden weight matrix
b_h is the bias for the hidden state
Tanh is the activation function

The architectures help in the forecasting demand, grid stability, and the potential faults, which enables proactive interventions that could enhance the grid stability and operational efficiency.

3.3.3 Multi-Objective Optimization in Power Systems

AI-driven methods accelerate the balancing multiple competing objectives, as a key challenge in modern power system operations [83]. The main issue of traditional model was their lack of capability in handling multiple approaches simultaneously. The ones like, cost minimization, emission reduction, and reliability maintenance, alongside with the handling complex constraints. AI-based approaches (e.g., revolutionary algorithms and reinforcement learning) can naturally optimize across multi-dimensional objective spaces, that can produce better solutions to deal with the real-world complexities of power systems.

3.4 Implementation Considerations and Challenges

Despite of the significant advantages that AI bring for the power systems, its deployment often comes with some challenges that must be addressed to ensure its successful implementation.

3.4.1 Data Quality and Availability

Suitable performance of AI is highly dependent on the quality and availability of data [84]. As mentioned before, for a proper forecasting, many AI applications require historical data that could be:

- Incomplete, due to lack of systemic collection
- Inaccurate, resulting from machine or human errors
- Low-resolution, as traditional measurement systems may not capture the high-frequency, synchronized data needed for advanced AI techniques.

This issue must be addressed before AI deployment. There are some ways that can be considered to overcome the mentioned issues [85]:

- Equipment upgrade for the high-resolution data collection
- Data cleaning and preprocessing pipelines implementation
- Using artificial data generation in the places where real-world data is insufficient or imbalanced injected data

3.4.2 Interpretability and Explainability

AI models known for their complexity (act as a black box), yet sometimes they need to be transparent and interpretable, especially when used in critical infrastructure [86]. Recent advancement in Explainable AI (XAI) address this issue though techniques like:

- Attention mechanism that identifies which input features most influence AI predictions [87]. It helps operators and regulators for having a better understanding of AI decision-making, most importantly, for critical conditions.
- Feature importance analysis, that highlights the features that have the most impact over the output.
- Symbolic regression, which attempts to express AI models in a more interpretable mathematical form.

3.4.3 Computational Requirements and Edge AI

AI methods vary widely in computational complexity, where deep learning methods mostly require high processing power for both training and real-time interference [88]. AI deployment directly in centralized cloud systems could bring some problems, like:

- High latency, that is problematic for the real-time grid operations
- Reliability issues, especially in the presence of communication failures

To address these issues, a promising solution is to use Edge AI [89], that intelligence is distributed across the power system. This architecture balances fast local responsiveness with the system-wide coordination. This approach will make application to be viable even in networks with limited communication bandwidth.

3.4.4 Regulatory and Organizational Challenges

In various scenarios, non-technical challenges pose greater barriers to AI adoption than the technical ones [90]. One Major issue is that existing regulations are built around traditional, rule-based control systems. AI-driven models which rely on probabilistic decision-making, may not be easily accommodated with these frameworks. One approach for this issue is to update policy and regulatory adjustments to ensure compliance and reliability. Another challenge is related to the data fragmentation. Various data gathering organization often manage their own isolated data sources. These entitles may be reluctant to share critical information, implementing AI over the whole system could become significantly complex. Lack of seamless data integration could damage the full picture needed for optimal decision-making process in AI models.

Rather than the regulatory and data concerns, organization resistance can also make the AI adoption slower than normal. Traditional operational frameworks have existed long enough for many power system professional, who spend years working within them, and AI-driven decision-making system adaptability with new recruits is the challenging section. It needs new training, policy shifts, and a change in mindset. Resistance against the changes (because of lack of trust in AI or concerns about the job security), can cause additional layer of difficulty in implementation.

3.5 Future Directions and Emerging Trends

With the advances o AI in the practical scenarios like power systems, several emerging trends poised to shape future progressions.

Hybrid AI methodologies, that merge multiple techniques, that benefit from their combined strength, are well-suited for these applications [91]. Notably, physics-informed neural networks (PINNs) integrate core power system equations directly into their learning structures [92]:

$$L_{total} = L_{data} + \lambda L_{physics} \quad (40)$$

L_{total} is the total outcome or the final loss
L_{data} represents the part of outcome that comes purely from data
$L_{physics}$ comes from the physics-based model
λ is the wight that controls the strength of reliance on the physics part

This formula depicts the AI-driven solutions adhere to fundamental physical laws while focusing on the generalize intricate patterns detection from the datasets, which enhances the performance even with limited training time and data.

Federated learning represents another significant trend, concentrating on collaborative model training without centralizing sensitive operational information [93].

$$\theta_{global} = (1/N) \times \sum \theta_i \quad (41)$$

θ_{global} is the global model that is the central result created after combination all the smaller models
N is the total number of local models
$\sum \theta_i$ represents the sum of all local models
θ_i is the local model for specific i

This decentralized approach effectively addresses privacy concerns and data ownership issues, that allows the comprehensive learning over the diverse operational conditions.

Digital twin technology embodies an ambitious and transformative direction, that AI-driven models continuously interact with the physical counterparts [94]. With the utilization of this virtual replicas, robust scenario testing, optimization, and system analysis without endangering the real-world equipment's due to out-of-control conditions. The dynamic, two-way information flow ensures the mutual enhancement and continuous refinement of both virtual and physical systems.

Quantum computing also holds significant promise in overcoming current computational barriers inherent for the complicated power system challenges [95]. These methodologies offer exponential computational acceleration for specialized problems, enables the resolution of previously unmanageable optimization scenarios.

By expansions and development of these methods, AI role will expand from supporting isolated system functions to orchestrating comprehensive interaction over different aspects of power systems (generation, distribution, and consumption).

4 AI Applications for Power Quality Enhancement, Monitoring, and Control: A Comprehensive Integration

Incorporation of AI into Power Quality (PQ) technologies has evolved the management practices in electrical distribution systems [96]. Different power systems issues like voltage sags, power factor correction, and real-time monitoring have seen significant improvement due to the utilization of AI-driven interventions. This section thoroughly explores these interconnected application, that offers the strength of AI methods to advance the power quality management using detailed mathematical frameworks, comparative analysis, and sophisticated optimization strategies.

4.1 Advanced AI Technologies

With numerous number of AI techniques, here a few beneficial ones are discuses alongside their strengths that they bring to the PQ management.

4.1.1 Neural Network Architectures

Neural networks can be implemented in various PQ applications, where it leverages specialized structures suited for various scenarios:

4.1.1.1 Feedforward Neural Networks (FFNN)

It features a straightforward architecture with the consideration of input, hidden, and output layers [97]. The related mathematical operation formula is:

$$y_j = f(\sum W_{ij} x_i + b_j) \quad (42)$$

x_i represents inputs
W_{ij} denotes weights
b_j indicates biases
f is an activation function

4.1.1.2 Long Short-Term Memory (LSTM)

This method has proven its high effective in predicting voltage sags by capturing temporal patterns [98]. The related mathematical equation to this methods follows as following:

$$c_t = f_t \times c_{t-1} + i_t \times \tilde{C}_t \quad (43)$$

c_t is the cell state at time t
f_t is the forget gate's output
i_t is the input gate's output
$\tilde{C}_t$ is the candidate cell state

4.1.1.3 Convolutional Neural Networks (CNNs)

It excels at extracting spatial patterns form PQ waveforms, which makes them ideal for the future extraction in disturbance classification tasks [99]. The formula is:

$$a_j^l = \sum\sum W_{mn}{}^k \times a_{(i+m)(j+n)}{}^{(l-1)} \quad (44)$$

a_j^l is the output of the k-th convolutional filter
$W_{mn}{}^k$ represents filter weights
$a_{(i+m)(j+n)}{}^{(l-1)}$ denotes input values within the filter window

4.1.2 Fuzzy Logic Systems

The utilization of the fuzzy logic controllers are widely for the PQ applications due to their ability to handle uncertainty and imprecise information effectively [100]. These controllers provide robust decision-making capabilities by resembling the human-decision making behaviour and linguistic rules rather than relying solely over mathematical models. The fundamental steps:

- Fuzzification: Numerical inputs conversion into fuzzy sets with defined membership values
- Rule Evaluation: Predefined rule bases utilization and applies fuzzy operators to assess system conditions.
- Aggregation: Output merging to generate a combined fuzzy response.
- Defuzzification: Aggregated fuzzy output conversion into a precise control action for the system application.

In PQ, especially for FACTS devices, fuzzy logic is instrumental in dynamically adjusting control parameters. This method typically processes inputs such as error and change in error to find an appropriate control action. The mathematical representation of this procedure is as follows [101]:

$$\text{IF } e \text{ is } E_i \text{ AND } \Delta e \text{ is } \Delta E_j \text{ THEN } u \text{ is } U_k \quad (45)$$

E_i, ΔE_j, U_k are the membership functions representing the linguistic variables for the system inputs and outputs
e is the error
Δe is the change in error
u is the appropriate control action

These rules help optimize the performance of PQ devices such as DVRs and DSTATCOMS by ensuring real time adaptive control.

4.1.3 Evolutionary and Swarm Intelligence Algorithms

For the matter of optimization and efficiency enhancements of PQ in control systems, evolutionary computing and swarm intelligence algorithms have been widely applied. By emulation of biological processes and natural behaviours, these methods try to solve complex optimization problems efficiently.

4.1.3.1 Particle Swarm Optimization (PSO)

It is a population-based optimization technique which is inspired by the behaviours of birds and fish [102]. This method tries to refine the potential solutions using the particle velocities update and positions according to the following mathematical equations:

Fuzzy logic controllers provide robust performance in power quality applications by managing uncertainty and imprecision. The basic fuzzy inference process involves:

$$v_i^{t+1} = w \times v_i^t + c_1 r_1 (pbest_i - x_i^t) + c_2 r_2 (gbest - x_i^t) \quad (46)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (47)$$

w is the inertia weight
c_1 and c_2 are acceleration coefficients
r_1 and r_2 are random values ensuring stochastic behavior
$pbest_i$ represents the personal best position of each particle
gbest is the best global position found by the swarm

this method has been widely used in PQ application, like, voltage sag detection, harmonic compensation, and optimal placement of FACTS devices.

4.1.3.2 Genetic Algorithms (GA)

That provides a robust optimization using the principles of natural selection, crossover, and mutation [103]. By iteratively evolve candidate solutions to achieve optimal system performance, they can be used in power systems quality control for various scenarios. Proportional-I ntegral (PI) tuning in FACTS devices is a common usage of this method. The goal of this optimization is to minimize the following fitness function:

$$J = \int (w_1 \times e(t)^2 + w_2 \times u(t)^2) dt \quad (48)$$

e(t) is the error signal between the desired and the actual value
u(t) is the control effort that is applied to the system
w_1 and w_2 are weighting accuracy for implementing the balance among accuracy and control energy

With the optimization of control parameters, GA ensures about the improvement in transient response, reduce overshoot, and enhanced stability in PQ devices. With the combination of this method with other AI techniques, the hybrid model is capable of enabling intelligent and adaptive power quality management, that improves the grid reliability significantly.

4.2 AI Techniques for Voltage Sag Prediction and Mitigation

Voltage sag is one the crucial issues of the power systems that needs to be addressed [104]. It's prediction and mitigation are vital for the power quality management, where AI-driven techniques provide advanced predictive analytics and control mechanism for this matter. Using these methods would enhance the grid stability and reliability with enabling the proactive responses to voltage disturbances.

4.2.1 Predictive Analytics for Voltage Sag

In the modern voltage sag prediction frameworks, multiple AI modules will collaborate together to perform specialized tasks, such as, feature extraction, classification, and time-series forecasting.

4.2.1.1 Feature Extraction and Selection

For the comprehensive voltage signal analyze, wavelet transform techniques are widely utilized [105] [177]; by breaking them down into multi-resolution components, they can capture the transient events effectively. The mathematical formula is:

$$WT_x(a,b) = (1/\sqrt{a}) \times \int x(t) \psi((t-b)/a) dt \quad (49)$$

a is the scale parameter for the frequency resolution control
b is the translation parameter shifting the function in time
ψ is the mother wavelet function

4.2.1.2 Classification Models

Support Vector Machines (SVMs) with Gaussian kernel function are common used for the power quality disturbances classification, that provides a high accuracy and robustness [106] [107]. The kernel function is as follows:

$$K(x_i, x_j) = \exp(-\|x_i - x_j\|^2 / 2\sigma^2) \quad (50)$$

With the decision function given by:

$$f(x) = \mathrm{sign}(\sum \alpha_i y_i K(x, x_i) + b) \quad (51)$$

x_i, x_j are feature vectors representing different data setpoints
$\|x_i - x_j\|^2$ is the squared Euclidean distance between the feature vectors
σ is a hyperparameter that controls the influence of a single training
α_i is the LaGrange multiplier
y_i is the class label for each training sample
$K(x, x_i)$ is the kernel function output
b is the bias term that helps to define the decision boundary in the feature space

4.2.1.3 Time-Series Prediction

The main privilege of LSTMs networks in forecasting voltage events are analyzing historical voltage data [108]. The LSTM cell state evolution is governed by:

$$h_t = o_t \times \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (52)$$

h_t is the hidden state at time t
W_c is the weight matrix governing state transitions
x_t is the input voltage data
o_t represents the output gate activation

4.2.2 Advanced FACTS Devices with AI Control

FACTS devices which are equipped with AI-based controllers are capable of real-time adaptive solutions for mitigating voltage sags using their adjustable reactive power [109]. AI could enhance the precision and efficiency of these controllers, ensuring a rapid response to any grid disturbances with variety of severity.

4.2.2.1 Static Synchronous Compensator (STATCOM) with AI Control

The operation of AI integration to STATCOM can be formulized as [110] [111]:

$$V_{PCC} = V_s - Z_s(i_L - i_{STATCOM}) \quad (53)$$

$$i_{STATCOM} = I_{mag} \times \sin(\omega t + \varphi) \quad (54)$$

V_{PCC} voltage at the Point of Common Coupling (PCC)
C is the capacitance of the DC-link capacitor
i_L is the load current drawn from power system
$i_{STATCOM}$ is current injected by the STATCOM to compensate for voltage sags and maintain the grid stability
I_{mag} is the magnitude of the compensating current, is adjusted by the AI model
ωt is the grid regular frequency term
φ is phase angle that is optimized by a neural network controller

4.2.2.2 Dynamic Voltage Restorer (DVR) with AI-Enhanced Sliding Mode Control

DVRs utilize advanced Sliding Mode Control (SMC) with AI-optimized procedures to compensate for voltage sags [112] [113]. The sliding surfaces for direct and quadrature-axis error compensates are defined as the following mathematical equations:

$$s_d = e_d + k_d \times \int e_d dt \quad (55)$$

$$s_q = e_q + k_q \times \int e_q dt \quad (56)$$

s_d, s_q sliding surface variables in the direct (d) and quadrature (q) reference frames
e_d, e_q voltage errors in the d-q frame
k_d, k_q control parameters tuned by the AI-based optimizer

In the AI-enhanced novel reaching law ensures rapid convergence of the sliding mode control:

$$u = -\eta \times |s|^\alpha \times \mathrm{sign}(s) \quad (57)$$

η is the adaption coefficient controlling the rate of system response
α exponent regulating the nonlinear behaviour of the reaching law
sign(s) is the sing function ensuring the correct switching behaviour

4.2.2.3 Unified Power Quality Conditioner (UPQC) with AI Integration

The UPQC is a multi-functional power quality device that uses both series and parallel compensators for the voltage sags, harmonics, and power imbalances mitigation simultaneously [114]. AI-based controllers could optimize its performance using the series Active Power Filter with adjusting the reference compensation voltage [115]:

$$V_{se\_ref} = V_{L\_ref} - V_s \quad (58)$$

V_{se_ref} is the reference compensation voltage injected by the series APF
V_{L_ref} is the desired load voltage after compensation
V_s is the measured distorted source voltage before compensation

4.3 AI-Driven Dynamic Power Factor Correction and Reactive Power Compensation

AI-based power factor correction plays a major role in enhancing power quality with reactive power compensation dynamically [116]. Neural network-driven approaches offer precise real-time adjustments, to ensure the system efficiency and stability improvement.

4.3.1 Neural Network-Controlled Shunt Active Power Filters

Shunt Active Power Filter (SAPFs) that are equipped with the neural network control operate, are one of the integration of AI and FACTS [117]. In this system harmonic currents will compensate to improve the power factor and mitigate distortions. The fundamental compensation principle is expressed as:

$$i_s = i_L - i_c \quad (59)$$

i_L represents the load current
i_c is the compensation current
i_s is the desired source current

Neural network can determine the desired source current i_s based on the fundamental power calculations [118]:

$$i_s = (2P/3V_s) \times \sin(\omega t) \quad (60)$$

$$i_c = i_L - i_s \quad (61)$$

By continuously adjusting the compensation current, neural network controls the P and Q which are active and reactive power respectively, to maintain the unity power factor of the grid.

4.3.2 Adaptive Reactive Power Compensation Algorithms

For the advanced reactive power compensation techniques, that incorporate instantaneous power theory enhanced using neural-network based optimization [119], there is a mathematical quantification for the active and reactive power as follows:

$$[p\ q]^T = [v_\alpha\ v_\beta;\ v_\beta\ {-v_\alpha}] \times [i_\alpha\ i_\beta]^T \quad (62)$$

p represents the instantaneous active power
q represents the instantaneous reactive power
v_α and v_β are the transformed voltage components
i_α and i_β are the transformed current components

The neural network has a vital role in optimizing the extraction of oscillating components (p and q), to enable the selective harmonic compensation while preserving the fundamental reactive power component required for the unity power factor correction.

The reference current required for compensation is computed as:

$$[i_{c\alpha}\ i_{c\beta}]^T = (1/(v_\alpha^2 + v_\beta^2)) \times [v_\alpha\ v_\beta;\ v_\beta\ {-v_\alpha}] \times [\tilde{p}\ q + \bar{q}]^T \quad (63)$$

Utilizing this approach enables the precise reactive power compensation while ensuring a unity power factor alongside with the harmonic distortion minimizing.

4.3.2 Distribution Static Compensators with Intelligent Control

D-STATCOM with the aid of AI based control strategies could demonstrate enhanced dynamic performance for the power factor correction and reactive power compensation [120]. AI-driven approaches have the capability of real-time adaption to system conditions, that improves the response time and stability. This compensating reactive power in D-STATCOM system is given by:

$$Q = V_s \times I_q = V_s \times |V_c| \times \sin(\delta) / X \quad (64)$$

Q is the compensating reactive power
V_s is the system voltage
I_q is the compensator current
δ is the compensating phase angle

Based on this equation, D-STATCOM can rapidly adjust its angle to maintain the power factor close to unity, in order to ensure that efficient power delivery and reduced losses.

The Adaptive Neural Fuzzy Inference System (ANFIS) integrates fuzzy logic and Artificial Neural Networks (ANNs) for the D-STATCOM improvement control [121]. With this hybrid approach, adaptability, learning capability, and robustness in handling non-linear system behaviour can be improved. The equation related to it can be quantified as:

$$y = \sum w_i f_i / \sum w_i \quad (65)$$

y is the ANFIS system output
w_i is the rule firing strength
f_i is the output of individual fuzzy rules
N is the total number of fuzzy rules

Faster repose and better transient performance compared to the traditional controllers, can be named as one of the benefits of this AI-driven controller.

4.4 Real-Time Power Quality Monitoring and Distributed Control

Rather than AI, integration of other technologies like Inter-of-Things (IoT) and edge computing has revolutionized the real-time power quality monitoring and control [122], that allows the system to lead to a more efficient and intelligent decision-making form of operation.

4.4.1 AI-Enhanced Disturbance Classification

Multi stage AI architectures for the disturbance classification with high accuracy, has been used in the modern power quality monitoring [123]. The procedure contains:

- Signal Pre-processing: Wavelet packet decomposition extracts key voltage wave fore features.
- Feature Selection: With the utilization of Information Gain Ratio (IGR), various features importance can be determined.
- Classification: Using combined multiple models (ensemble method) could enhance the classification accuracy.

The mathematical formulation for IGR in feature follows as [124]:

$$IGR(A) = IG(A) / SplitInfo(A) \quad (66)$$

IG(A) represents the information gain for attribute A
SplitInfo(A) measures the entropy of attribute A

4.4.2 Distributed Edge Computing Architecture

With the utilization of edge computing [125], localized AI processing, distributing computational tasks across multiple nodes, enables. This technology matters as it runs a lightweight neural network [126]:

$$y_i = \varphi(W_i x_i + b_i) \quad (67)$$

y_i is the output at the ith edge node.
φ is the activation function
W_i and x_i represent the weight matrix and input vector, respectively

4.4.3 IoT-Integrated Monitoring System

A hierarchical data processing approach is utilized in the IoT-driven monitoring framework [127] [128]:

- Sensor Layer: Captures high-resolution voltage and current waveforms
- Edge Processing Layer: Performs the local analyzed with lightweight neural networks
- Fog Computing Layer: Manages regional coordination and preliminary analytics
- Cloud Computing Layer: Conducts system-wide alongside with the in-depth analysis

4.5 Optimization Strategies for Power Quality Systems

The main usage of AI-driven systems is the improvement in the power quality, enabling precise parameter tuning and efficient system configuration [129].

4.5.1 Multi-Objective Optimization for FACTS Device Placement

The suitable placement of FACTS devices employs multi-objective optimization to balance conflicting objectives [130] [131]:

$$\min/\max F(x) = [f_1(x), f_2(x), \ldots, f_n(x)] \quad (68)$$

Common objective functions include:

f_1 is the voltage sag mitigation
f_2 is the installation and operational costs
f_3 is the power system losses

The Non-Dominated Sorting Genetic Algorithm II (NSGA-II), as proposed by Nishad et al. (PDF) An ANN-Based Approach for Optimal Placement of ... - ResearchGate, optimized the placement of Unified Power Quality Conditioner (UPQC) devices, achieving a 28.7% reduction in installation costs while maintaining 96.3% effectiveness in voltage sag mitigation.

4.5.2 Metaheuristic Algorithms for Controller Parameter Optimization

Metaheuristic optimization techniques fine-tune controller parameters in PQ systems for the enhanced performance and reliability.

4.5.2.1 Teaching-Learning-Based Optimization (TLBO)

In this approach, teacher and student are used for the TLBD algorithm models the learning process [132]:

$$X_{new} = X_i + r \times (X_{teacher} - (T_F \times M_{mean})) \quad (69)$$

X_i represents the ith candidate solution
$X_{teacher}$ is the best-performing solution
M_{mean} is the population mean
r is a randomly generated number
T_F is the teaching factor

This approach was applied to power quality control in Real-Time Voltage Sag Detection and Classification for Power Quality, demonstrating robust performance improvements.

4.5.2.2 Grey Wolf Optimizer (GWO)

This algorithm is inspired by the social hierarchy of grey wolves, the GWO algorithm upgrades its position using [133]:

$$D = |C \times X_p - X| \quad (70)$$

$$X(t+1) = X_p - A \times D \quad (71)$$

X is the current position of a grey wolf in the research space
X_p is the position of the prey, or the current best solution
D represents the distance vector between the prey and current wolf
A and C are the vector coefficient

4.5.3 Deep Reinforcement Learning for Adaptive Control

Deep Learning Learning (DRL) could enable the adaptive control that allows an agent to learn optimal control strategies through interactions with its environment [134]. This process contains selecting actions based on states, receiving rewards, and adjusting actions by considering experience.

The State-Action-Reward-State-Action (SARSA) algorithm, updates the Q values each time through the following equation [135]:

$$Q(s, a) = Q(s, a) + \alpha \times [r + \gamma Q(s', a') - Q(s, a)] \quad (72)$$

Q(s, a) is the current estimate of the Q-value for state
α is the learning rate
r is the reward received after taking action
γ is the discount factor
s', a' are the next state and action

Utilization of this algorithm, helps the controller adaptively improvement and actions related to that which are based on real-time outcomes.

For more advanced scenarios, Deep Q-Networks (DQN) are used to approximate the Q-values using the deep neural network, particularly when the state or action space has large amount of numbers to be handled with traditional methods [136]:

$$Q(s, a; \theta) \approx Q^*(s, a) \quad (73)$$

θ contains various parameters (e.g., weights and biases) of the neural network
Q* is the optimal Q-value that network aims to approximate

This method uses past experience (the technique called experience replay) which stores and randomly samples the past interactions to improve learning stability and efficiency.

4.6 Integrated System Performance and Comparative Analysis

The ultimate combination of AI-driven sag reduction, dynamic power factor correction, and real-time monitoring enables a comprehensive and intelligent power quality measurement system. This combined method delivers robust performance across several critical demonstrations, aiding the grid stability and reliability to improve under various operating conditions.

4.6.1 Performance Evaluation Metrics

For the model effectiveness evaluation, multiple performance indicators are employed across three core functionalities [137] [138] [139]:

- Voltage Sag Compensation Effectiveness:
  Response Time: Duration between sag detection and mitigation action
  Voltage Total Harmonic Distortion (THD): Degree of waveform distortion after compensation